Paths in a product type are pairs of paths: the canonical function from x = y to pairs of component paths is an equivalence

The Univalent Foundations Program

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Editorial note. The counted statement is the author theorem labelled thm:path-prod (source lines 1249-1251); the proof span 1259-1284 is the author's proof environment, retained line for line.

Editorial note. The equation cited by the statement is retained as context, so both citations resolve inside the composed document.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the characterization of paths in cartesian products as pairs of paths. Only the counted theorem and its author proof are retained; the follow-up naming of the inverse and the uniqueness-principle remark remain uncounted in the source.

Allowed standard prerequisites (editorial).

- Cartesian product types with their induction principle: to construct or prove something about arbitrary elements it suffices to consider pairs.
- The action of the projections on paths: ap of the first and second projection applied to a path in the product.
- The induction principle for identity types (path induction) with its conversion rule.
- The notion of quasi-inverse of a function.

For any elements $x, y : A \times B$ and a path $p : x =_{A\times B} y$, by functoriality we can extract paths $\mathsf{pr}_1(p) : \mathsf{pr}_1(x) =_A \mathsf{pr}_1(y)$ and $\mathsf{pr}_2(p) : \mathsf{pr}_2(x) =_B \mathsf{pr}_2(y)$. Thus, we have a function

$$(x =_{A\times B} y) \to (\mathsf{pr}_1(x) =_A \mathsf{pr}_1(y)) \times (\mathsf{pr}_2(x) =_B \mathsf{pr}_2(y)). \tag{0.0.1}$$

Theorem 0.0.2. *For any x and y, the function* (0.0.1) *is an equivalence.*

Proof. We need a function in the other direction:

$$(\mathsf{pr}_1(x) =_A \mathsf{pr}_1(y)) \times (\mathsf{pr}_2(x) =_B \mathsf{pr}_2(y)) \to (x =_{A\times B} y). \tag{0.0.3}$$

By the induction rule for cartesian products, we may assume that x and y are both pairs, i.e. $x \equiv (a, b)$ and $y \equiv (a', b')$ for some $a, a' : A$ and $b, b' : B$. In this case, what we want is a function

$$(a =_A a') \times (b =_B b') \to \big((a, b) =_{A\times B} (a', b')\big).$$

Now by induction for the cartesian product in its domain, we may assume given $p : a = a'$ and $q : b = b'$. And by two path inductions, we may assume that $a \equiv a'$ and $b \equiv b'$ and both p and q are reflexivity. But in this case, we have $(a, b) \equiv (a', b')$ and so we can take the output to also be reflexivity.

It remains to prove that (0.0.3) is quasi-inverse to (0.0.1). This is a simple sequence of inductions, but they have to be done in the right order.

In one direction, let us start with $r : x =_{A\times B} y$. We first do a path induction on r in order to assume that $x \equiv y$ and r is reflexivity. In this case, since $\mathsf{ap}_{\mathsf{pr}_1}$ and $\mathsf{ap}_{\mathsf{pr}_2}$ are defined by path induction, (0.0.1) takes $r \equiv \mathsf{refl}_x$ to the pair $(\mathsf{refl}_{\mathsf{pr}_1 x}, \mathsf{refl}_{\mathsf{pr}_2 x})$. Now by induction on x, we may assume $x \equiv (a, b)$, so that this is $(\mathsf{refl}_a, \mathsf{refl}_b)$. Thus, (0.0.3) takes it by definition to $\mathsf{refl}_{(a,b)}$, which (under our current assumptions) is r.

In the other direction, if we start with $s : (\mathsf{pr}_1(x) =_A \mathsf{pr}_1(y)) \times (\mathsf{pr}_2(x) =_B \mathsf{pr}_2(y))$, then we first do induction on x and y to assume that they are pairs (a, b) and (a', b'), and then induction on $s : (a =_A a') \times (b =_B b')$ to reduce it to a pair (p, q) where $p : a = a'$ and $q : b = b'$. Now by induction on p and q, we may assume they are reflexivities refl_a and refl_b, in which case (0.0.3) yields $\mathsf{refl}_{(a,b)}$ and then (0.0.1) returns us to $(\mathsf{refl}_a, \mathsf{refl}_b) \equiv (p, q) \equiv s$. □